If the Diego Garcia South 01:02:53 signal were the impact (most probable source 37.1 S 89.9 E, impact 00:21:08 UTC): the data held at every receiver when its sound should arrive

needed coupling 0.00026 (1 times the F-35A value 2.1e-4; P_D = 0.5 at Diego Garcia South).
ranking track: rank 1 by ln LR (-1.63); implied source there 37.02S 90.02E, impact 00:21:07.9.



Diego Garcia North, (IMS H08N)

No recording held. Predicted arrival 01:04:38 to 01:05:40 UTC. no recording obtained (IMS data not released to the project).

Perth Canyon logger 3376, raw recording

No recording held. Predicted arrival 00:41:02 to 00:48:49 UTC. no record overlaps the window (records are 307 s every 15 min, alternating slots).

Portland logger 3274, raw recording

No recording held. Predicted arrival 01:07:32 to 01:12:19 UTC. no record overlaps the window (records are 307 s every 15 min).

OHASISBIO NEAMS

No recording held. Predicted arrival 00:30:31 to 00:36:00 UTC. instrument lost on recovery: no data exist (Royer 2016 Table 1).

OHASISBIO WKER, Royer spectrogram, digitised

No recording held. Predicted arrival 00:49:35 to 01:00:58 UTC. no digitised Royer panel covers the window; raw data not held.

OHASISBIO NCRO, Royer spectrogram, digitised

No recording held. Predicted arrival 00:55:54 to 01:06:44 UTC. no digitised Royer panel covers the window; raw data not held.

Time (UTC, 8 March 2014). Orange band: predicted arrival window (95 %). Black line: the signal that defines the candidate. Purple line: expected level of the candidate if it were the impact; dotted purple: level for detection probability 0.5. Grey dashed lines: matched-filter thresholds for p = 0.3, 0.1, 0.05 (light to dark). Teal: Royer's picked arrivals (filled triangle and dotted line: his Table 5; open triangle: the arrival marks on his figures).

(i) Each row is one receiver. The orange band shows when sound from the candidate source would arrive at that receiver. A grey row means that we hold no recording for that time. The purple line shows how loud the candidate would be if it were the impact, with the coupling that it needs. If the purple line is high and the trace shows nothing above the dotted purple line, the receiver should have detected the sound and did not. Plots marked digitised are copies of published figures, not recordings. These are model predictions. The sound-loss model and the coupling are not calibrated.

(ii) Source: run J search-geometry sample (waypoints stratum seeds 1-8 + northern supplement; 00:19 Held Out, p 0.5) conditioned on the observed arrival (Gaussian kernel, sd from c and 3.6 s pick error); ESS 4860. Window = 2.5-97.5 % of predicted arrival incl. c and pick errors; plots window +/- 2 min. Level: E 10.70 GJ, tau90 0.2 s (sample rows conditioned on the observed arrival); SNR = peak 1 s 5-40 Hz level re 14-day background median, TL from the arc grid (IMOS: RAM-corrected table imos_tl_ram_arc.csv; IMS: KRAKEN - RAM delta, +4.8 dB triad gain), P_D = block 7 logistic fit, M1b, alpha 0.05 (IMS: proxy noise). M1b = envelope matched filter (18-template KRAKEN bank on IMOS, station templates on panels); p = (k+1)/(n+1) against 55 s window maxima of the logger's own 1-15 March background (3315 <- 3376, 3275 <- 3274 stand-ins) or, for panels, IMOS 3376 through the plot emulator; p for the window maximum is not corrected for window length. Kadri Fig. 9 panels: digitised plots, 20 Hz, nothing above ~7 Hz trusted; Kadri times: text time 00:52:03, caption time 00:54:30. Royer (2016) spectrograms: colour->dB inversion (+/-1.5 dB), band 5-40 Hz; picks from Table 5 (H01W figure labels run 13.6-16 s later than the table, unresolved). PREDICTIVE, NOT CALIBRATED; caveats C1 (slope misfit +10.4/+7.4 dB/octave at H01W/H08S) and C2 (coupling not calibrated; F-35A anchor 2.1e-4).